HERMES

# MERCURY

A free software HF modem and reliable data link

Email, files and messages, from below −10 dB SNR to 3.1 kbit/s

**Rhizomatica** • **the modem of HERMES** • **October 2026**

github.com/Rhizomatica/mercury • hermes.radio



# Outline

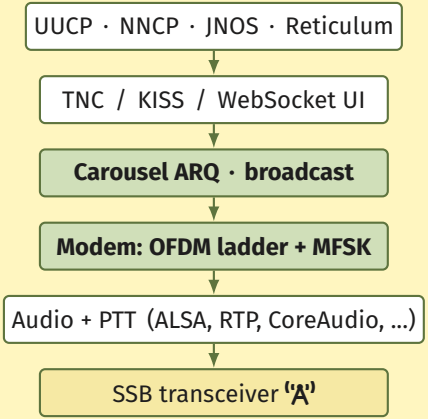
1. Mercury in HERMES
2. Modulation and coding modes
3. MFSK for weak-signal operation
4. Carousel ARQ
5. Formal methods: TLA+ and TLAPS
6. Testing and channel simulation
7. Integration and platforms
8. Outlook



# MERCURY IN HERMES

# Overview

- The HF software modem of **HERMES**: it turns an SSB radio into a data link for email, files and messages
- Hands the application a **TCP connection**: a VARA-compatible TNC (control 8300, data 8301), KISS and broadcast on 8100
- A clean **rewrite in C** (v2, since March 2026); the C++ v1 lives on as `mercuryv1`
- **Free software**: GPL-3.0-or-later; vendored codec2 subset LGPL-2.1
- Built by Rhizomatica for communities beyond the Internet, first in the Amazon rainforest



# 2026 in figures

**1109** commits on trunk since the C rewrite (2026-03-05)

**496** commits since 1 September

**5** releases since September (1.9.15 → 1.9.19)

## Platforms

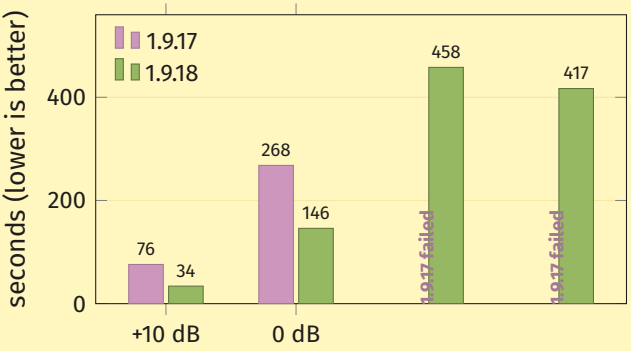
Linux (amd64, arm64, armhf, i386), Raspberry Pi 4/5, Windows (signed), macOS (universal, notarised); **iOS and Android coming soon**

## Releases

**1.9.18** (12 Oct): the carousel data plane
**1.9.19** (13 Oct): MFSK centred on 1500 Hz, protocol v2, JNOS 2.0

Source: git history of `mercuryv2`; docs/RELEASE-1.9.18.md, docs/RELEASE-1.9.19.md

# Release 1.9.18 compared with 1.9.17: a 2 KB transfer




- Two real daemons through codec2 `ch`, 2074 bytes
- **2× faster** on a good link
- Links down to −10 **dB** SNR now complete
- On air, 8 KB each way: 485/509 s → **312/238 s**
- At 3 % power, 2 KB: 1310/1266 s → **832/755 s**

Source: docs/RELEASE-1.9.18.md



# MODULATION AND CODING MODES

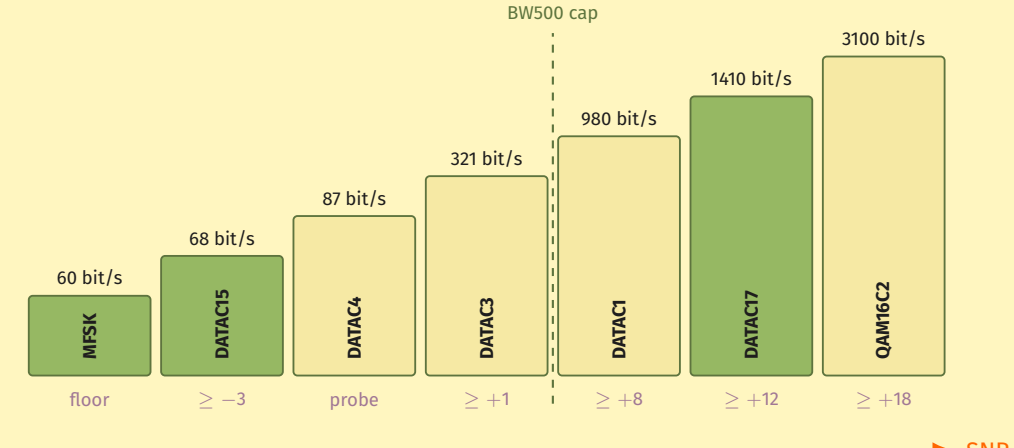
# Mode overview

| Mode | Origin | BW, Hz | Payload, B | bit/s | FEC (n,k) | Role |
|---|---|---|---|---|---|---|
| MFSK | Mercury | 1000 | 98 | ∼60 | LDPC (1600,800) | rung 0 (floor) |
| DATAC15 | Mercury | 200 | 30 | 68 | (768,256) | rung 1 |
| DATAC4 | codec2 | 250 | 54 | 87 | (1472,448) | rung 2 |
| DATAC3 | codec2 | 500 | 126 | 321 | (2048,1024) | rung 3 (BW500 top) |
| DATAC1 | codec2 | 1700 | 510 | 980 | (8192,4096) | rung 4 |
| DATAC17 | Mercury | 2100 | 1180 | 1410 | (15936,9456) | rung 5 |
| QAM16C2 | codec2 | 2100 | 1213 | 3100 | (16200,9720) | rung 6 |
| DATAC16 | Mercury | 200 | 14 | 42 | (640,128) | ARQ control |
| DATAC0/13/14 | codec2 | 200–500 | 3–14 | 58–291 | — | broadcast, tests |

green = designed by us: three of the seven rungs, and the control mode

Source: docs/MODES.md

# Adaptive mode ladder: seven rungs from 60 to 3100 bit/s




start rung chosen by the SNR the caller measured (dB); then link adaptation by measured goodput

Source: datalink_arq/carousel.c (car_start_level), docs/CAROUSEL-SPEC.md §15

# OFDM modes designed by the project

## Robust: DATAC15 and DATAC16

- Rate-1/3 LDPC (`H_256_768_22`), 200 Hz
- DATAC15: 30 bytes, 3 carriers, the first OFDM rung
- DATAC16: 14 bytes, calls, polls and ACKs
- 70/100 frames at −7 dB on MPP fading

## Fast: DATAC17, on codec2's QAM16C2 code

- 2.1 kHz, about 1.2 KB a frame
- codec2's QAM16C2: 16-QAM, (16200,9720), 3100 bit/s
- DATAC17: QPSK on that codeword, 1410 bit/s
- DATAC17 97/100 at +10 dB (MPP)

## Accuracy of link adaptation

Per-mode **SNR calibration** (QAM16C2 read 9 dB low); a 16-QAM estimator above 12 dB; the **TX filter** no longer cuts our outer carriers (6.5 dB); **FFT-correlation** acquisition, 2.2× cheaper

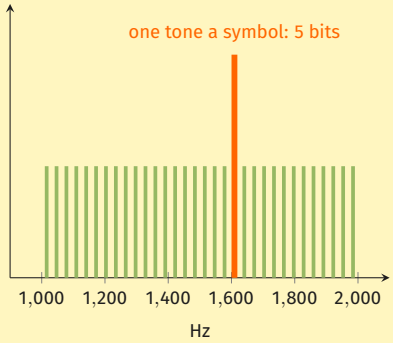
Source: docs/MODES.md, docs/OTA-PHASE-A-SNR-CALIB.md; PRs #163, #337, #350



# MFSK FOR WEAK-SIGNAL OPERATION

# MFSK: a non-coherent waveform below the OFDM modes

- Mercury's own modem: **32-FSK on OFDM subcarriers**, non-coherent
- 8 kHz, Nfft 256: tones 31.25 Hz apart, 40 ms symbols (25 baud)
- Centred on **1500 Hz** since 1.9.19: tones 1015.6–1984.4 Hz
- Preamble *and* postamble: a burst is found from either end
- LDPC N = 1600; five rates ported, **rate 1/2 ships** (98-byte frames, 13.5 s)
- Reducing the code rate to 1/16 gains only 2.1 dB: sensitivity is limited by synchronisation

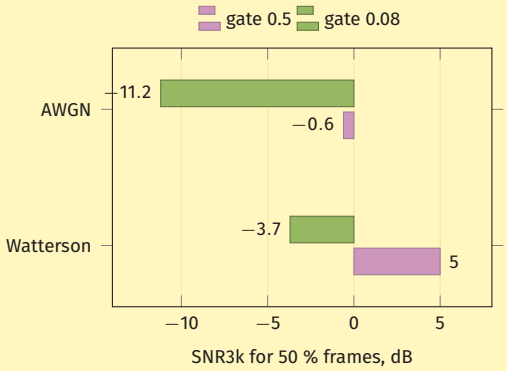



The 32 tones, 1015.6–1984.4 Hz

Source: docs/MFSK-PORT.md, modem/modem_mfsk.c, docs/RELEASE-1.9.19.md

# Sensitivity limited by synchronisation, not by coding

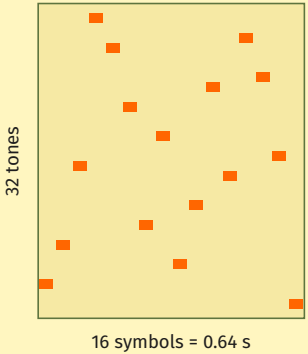
- All five code rates gave **identical** frame error rates: frames near the threshold never reached the decoder
- The sync accept gate was 0.5, which demands SNR per symbol $\geq 0$ dB
- At **0.08**, with false anchors released:
  - AWGN 50 % point: $-0.6 \rightarrow -11.2$ **dB**
  - Watterson 2 ms / 1 Hz: $+5.0 \rightarrow -3.7$ **dB**
- Dual-ended acquisition contributes about 2 dB; codeword interleaving adds 19 % throughput at the margin




Source: docs/MFSK-PORT.md

# Pattern acknowledgements: 0.64 s of airtime

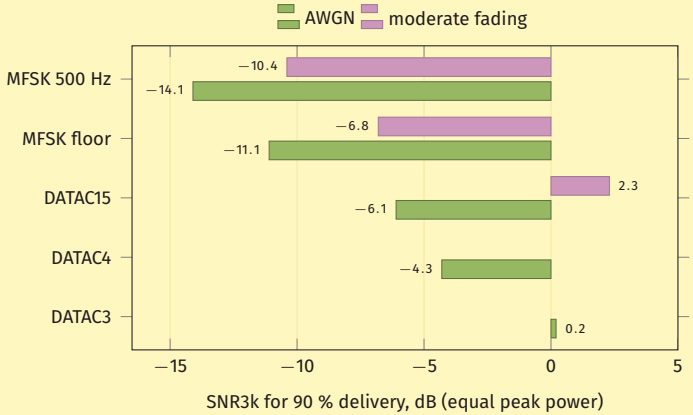
- An acknowledgement as a **Welch-Costas pattern**: 16 MFSK symbols, **0.64 s**
- 5.8× less airtime than a 3.74 s DATAC16 frame
- Detected every time down to about −9 dB on fading and −22 dB on AWGN
- **0 false alarms** in 30 trials down to −26 dB; 0 in 2 h of noise after the half-subcarrier search
- Frequency search over ±78 Hz: an IC-7100 and an sBitx went from 0/5 to 17/17 patterns on air




(illustrative layout: one tone per symbol, no repeats)

Source: docs/MFSK-PORT.md, README.md; PRs #160, #299

# Sensitivity of MFSK compared with the OFDM modes




- MFSK operates **5 to 9 dB** below DATAC15
- Against modem73's coherent RFDM modes: 2.7 dB better per bit on AWGN, a tie under fading
- On air: 1 KB over UUCP at **2 % power** in 533 s

Sim, 100 trials a point, Watterson model, 2026-10-15 (PR #390); docs/MFSK-PORT.md; docs/RELEASE-1.9.19.md

# Under review: 500 Hz MFSK and sample-clock tracking

## Mode 103 for 500 Hz filters (PR #390)

- The 1 kHz MFSK cannot be decoded behind a 500 Hz filter
- 32 tones 15.625 Hz apart, 80 ms symbols, 26 s bursts
- 90 %: −14.1 **dB** AWGN, −10.4 dB moderate, behind the filter too
- On air at the bench: −14.2 dB (sim −14.1)

## Three receiver fixes, all modes

- Dial offset to an eighth of a subcarrier
- Pattern search in half-subcarrier steps (removes a 3–4 dB loss)
- LLR noise from the tones' own bins
- Payload read at a **searched sample-clock ratio**: decodes from 0.98 to 1.02 (was ±0.2 %)
- Carrier offset as a top-level status field

Source: PR #390 (draft), measurements 2026-10-15/16



# CAROUSEL ARQ

# Motivation for a new ARQ

## Previous design: stop-and-wait

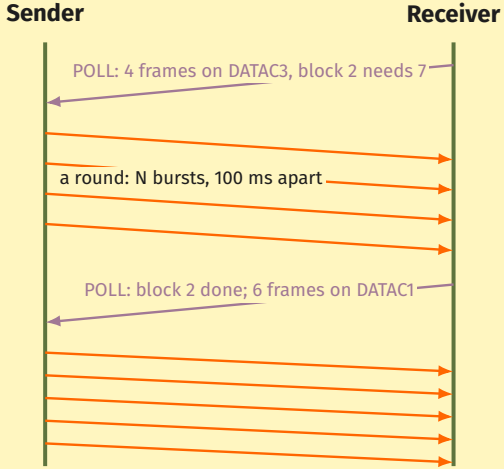
- SNR-driven: one frame, one ACK, one turnaround
- Rungs chosen from SNR estimates that were wrong per mode
- Lost ACKs on the weak reverse path stalled the session
- Sim NVIS: **0 of 20** runs completed

## Carousel design

- **Receiver-driven**: the receiver requests what it still needs
- **Erasure-coded**: any $k$ of the pieces rebuild a block, so no retransmission of a particular frame
- No per-frame ACKs, no sequence numbers
- Rungs chosen by **measured goodput**
- Merged 3 Oct (#322), the only data plane since 11 Oct (#364)

Source: docs/CAROUSEL-ARQ.md, docs/CAROUSEL-REPORT.md

# Carousel operation




- Systematic Cauchy **Reed-Solomon** over GF(256), blocks of up to 96 pieces
- **23–24 byte pieces**: four fit one MFSK frame
- Up to 8 blocks open; each ends in a seed-keyed CRC-32 and fails closed
- Four consecutive losses mark a rung unusable; it is probed again after 60 s, the interval doubling up to 8 min

Source: docs/CAROUSEL-SPEC.md, docs/CAROUSEL-ARQ.md

# Channel access, connection set-up and versioning

## Single timing master

The caller owns every timer; the callee keys only in its slots. **0 collisions** in 960 two-way deep-fade sim runs (#351)

## Connection at low SNR

After two unanswered DATAC16 CALLs, the CALL alternates with MFSK. The ACCEPT names the start rung and carries the callee's nonce

## Session binding

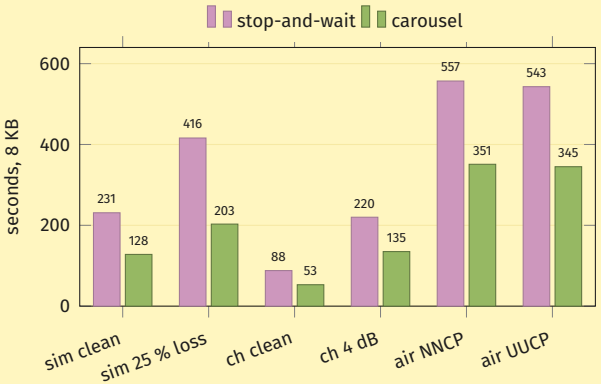
Seed = CRC16(session id, nonce, both calls), mixed into every frame's CRC. Protocol v1 = 1.9.18, v2 = 1.9.19: a CALL of another version is logged, not answered

## Listen before talk

Every keying path waits for a clear channel; the busy detector is opt-in (off by default)

Source: docs/CAROUSEL-TURNS.md, docs/CAROUSEL-SPEC.md; PRs #317, #351

# Results: simulation, modem bench and on-air tests




- **35–45 % shorter** transfer times in every measurement
- Simulated NVIS: stop-and-wait never completed; the carousel did
- At **2 % power** trunk never connected; the carousel delivered in 632–932 s
- On air: mean of two runs in each direction, sBitx test bench

Source: docs/CAROUSEL-REPORT.md, docs/CAROUSEL-ARQ.md



# FORMAL METHODS: TLA+ AND TLAPS

# Formal models of the protocol layers

| Model | What it proves | Checked |
|---|---|---|
| **CarouselBlocks** | InOrder, Integrity, RetireSafe, NoPhantom: delivery in order, nothing invented, nothing retired early | TLC up to **63.4 M states**; an adversary config at 12.1 M |
| **Liveness** | Completes: every block is delivered, under a lossy adversary | 2.58 M states, 14.2 h |
| **CarouselProofs** | Spec $\Rightarrow$ $\Box$Safety for any number of blocks | **TLAPS: 823 obligations** (Z3, Zenon) |
| **CarouselTurns** (timed) | NoOverlap: the two stations never key together; MNotShutOut | 18.5 M states; MFSK config 27.1 M |
| **RungAgreement** | Agree, Wasted, Delivered: both ends agree on the rung | 1458 states; 5592 at 4 rungs |

Source: specs/carousel, specs/turns, specs/rungs (README files)

# Validation of the models

## Mutation testing

- Turns: five mutants (no grid, late, long, blind, reuse) each violate NoOverlap
- Rungs: five more (no request, no bind, deaf, unnamed, previous rung)
- Blocks: the no-check config violates Integrity, as it must
- Before the integrity work, the model found **two real bugs**, 9 steps each

## Conformance of the implementation

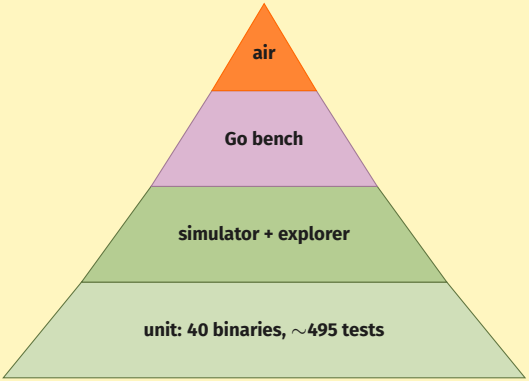
- `car_explore` runs the **real** `carousel.c` under every set of up to 3 losses or sensing failures, plus spurious BREAKs and corrupt frames
- **19 765 runs in 1.5 s** in CI; 9.09 M in the long sweep
- Overrun and slot oracles in the simulator check the timed model's assumptions

Source: specs/, tests/sim/car_explore.c; PRs #345–#347



# TESTING AND CHANNEL SIMULATION

# Test framework




- **Unit**: modems, LDPC, carousel, ARQ, TNC, JSON, rings, channel model
- **Simulator**: two state machines on a virtual clock; A/B benches on 120 seeds
- **Go bench**: two real daemons over FIFO, socket or RTP audio through `ch` or Watterson; two different builds can be paired
- **On air**: sBitx, IC-7100 and FT-710 stations, UUCP / NNCP / JNOS transfers
- **One framework**: no second copy of test data, no tool CI cannot run

Source: tests/Makefile, tests/integration, tests/sim

# Continuous integration on every pull request

## Builds

Linux amd64, arm64, armhf; Windows x64 (cross); macOS arm64; the Fyne GUI on Linux and Windows

## Tests

Unit tests on amd64 and **i386**; the Go integration bench

## Sanitizers

ThreadSanitizer on the unit tests; AddressSanitizer + UBSan on the unit tests and a full integration session

## Additional tooling

libFuzzer targets for the ARQ header, frame header, KISS, arithmetic and WebSocket JSON parsers; signed and notarised release pipeline

Source: .github/workflows, docs/SANITIZERS.md, docs/FUZZING.md

# Channel model compliant with ITU-R F.1487

- Watterson model with the **Gaussian Doppler spectrum** of **ITU-R F.1487** / **CCIR 520-2**: spread = $2\sigma$
- Named presets `ccir-good/moderate/poor/flutter` and ten `itur-*` (low to high latitude, NVIS)
- An exact delay line; the former Butterworth presets are retained for reproducibility
- The former “poor” preset faded **twice as fast** as CCIR Poor: 1.007 against 0.515 fades per second

## Acceptance tests in the CI suite

| | Measured | Theory |
|---|---|---|
| Power beyond $3\sigma$ | 0.273 % | 0.27 % |
| Power beyond the spread | 4.68 % | 4.55 % |
| Fades below $-10$ dB, $s^{-1}$ | 0.5067 | 0.5072 |

Reference values: the Gaussian spectrum and Rice’s closed-form level-crossing rate. Records of 900 s with fixed seeds.

## Cross-validation with MATLAB

Every preset compared with the R2026a Communications Toolbox (`stdchan`, `comm.RayleighChannel`): **26 of 26 checks passed**; a second run reproduced the results byte for byte.

Source: PR #379 (Joseph Freivald, approved): docs/CHANNEL-MODEL.md, tests/common/test_watterson.c

# Independent measurement tools

## Channel simulators

- codec2 `ch`: AWGN and MPP
- Our Watterson model (above)
- AE4JY's **PathSim**, calibrated against ours to $\pm$0.2 dB
- OpenARQ's **skywave** harness, with a Mercury adapter

## Probes and sweeps

- Pattern, acquisition and burst-train probes
- MFSK channel sweeps by file, through any channel tool
- An opt-in RX audio dump: a frame missed on air can be decoded offline
- On-air recordings with noise added offline: the bench's own fringe curve

Source: utils/, docs/MFSK-PORT.md; PR #389



# INTEGRATION AND PLATFORMS

# Supported applications

### Mail and files

UUCP is the HERMES default (`-Y` probe race fixed upstream); NNCP works too; JNOS 2.0 BBS-to-BBS and telnet with no IP on air

### Networks

Reticulum over KISS on the broadcast port; broadcast files by a RaptorQ fountain, wire-compatible with hermes-broadcast

### Encryption (in review)

Optional Noise_KK sessions (X25519, ChaCha20-Poly1305), XChaCha20 broadcast files; off by default (PR #306)

Source: docs/JNOS.md, docs/RETICULUM.md, docs/BROADCAST-FILE.md; PR #306

# Interfaces and platforms

- **Our own WebSocket server**: 28 820 vendored GPL-2.0-only lines replaced by about 1500 of ours (RFC 6455, TLS through OpenSSL)
- Status: modes, the peer's view of our signal, frequency, **carrier offset**
- **TUNE**: a 1 kHz tuning carrier, hard 60 s unkey deadline
- Audio: ALSA, Pulse, CoreAudio, AAudio, WASAPI, DirectSound, **RTP** to hermes-radio-daemon
- PTT: Hamlib, serial RTS, CM108 GPIO, HERMES shared memory
- **Fyne GUI** with the engine linked in
- **Windows**: Authenticode-signed (Certum)
- **macOS**: universal x86_64 + arm64, static Hamlib, notarised
- **iOS and Android**: coming soon, the modem and its GUI on the phone
- **Raspberry Pi 4**: idle receive at 28–30 % of one core

Source: README.md, docs/WEBSOCKET_STATUS.md, docs/WINDOWS-SIGNING.md, docs/MACOS-UNIVERSAL.md; PR #383



# OUTLOOK

# Future work

### Mercury on iOS and Android

The modem and its GUI on the phone: coming soon

### Optional encryption

Encrypted sessions on the carousel plane, off by default (Noise_KK, PR #306)

HERMES

# THANK YOU

- github.com/Rhizomatica/mercury
- hermes.radio
- docs/ in the repository: modes, carousel, specs, channel model

Mercury is free software (GPL-3.0-or-later).

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